

An Integrated Framework for Flood Risk Assessment

by

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Scanning the banks of a river in Pune, Maharashtra, India.

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Introduction

Background and Motivation

Pune, a city in Maharashtra, has experienced substantial land-use change, with a 322% increase in built-up area from 1995 to 2025, representing one of the highest rate of urban expansion in India [KM25]. This growth has been accompanied by increased flooding, frequently associated with the Mula and Mutha rivers, which drain the city's central catchment [DV24, Dwi24, Nam19].

Urban expansion alters natural hydrological processes by increasing impervious surface area, thereby reducing infiltration and increasing runoff entering rivers. In addition to urbanisation, recent infrastructural developments, particularly the Pune Riverfront Development, have modified the Mula-Mutha river system at a more fundamental level. These interventions involve channelisation, embankment construction, and modification of natural floodplains [Tan25, Jad26] (**Figure1**²). To understand the impact of these modifications, it is necessary to examine how the natural structure of riverbanks influences flood risk.

This perspective highlights that flood risk is governed by both **extrinsic** factors, such as rainfall and urbanisation, and **intrinsic** factors, relating to the **geometric-geomechanical properties**³⁴⁵ (GGP) of the riverbanks. While extrinsic drivers are widely studied in urban hydrology, intrinsic riverbank stability is often treated separately within geotechnical frameworks, despite its influence on failure risk, a key gap this study seeks to address.

Inspired by the **Human Development Index**⁶ (HDI), which combines socioeconomic indicators into a single measure, we develop an analogous framework for river systems. The proposed **Bank Stability Index**⁷ (BSI) aims to combine key intrinsic GGP into a single quantitative measure of riverbank stability, and hence flood vulnerability.

²**Figure1**, along with all other tables and figures, can be found in Appendix D and Appendix E respectively.

³ Technical terms are bolded, with definitions provided in footnotes on the page where they first appear, as well as in Appendix A for reference. These standard definitions are primarily from the relevant Wikipedia pages.

⁴ **Geomechanical-Geometric Properties (GGP)**: Combined set of soil mechanical properties (e.g., cohesion, internal friction, porosity) and geometric characteristics (e.g., slope angle, height, width) that govern the behaviour of a geological structure.

⁵ A complete list of abbreviations is given in Appendix C.

⁶ **Human Development Index (HDI)**: A composite index used to measure and compare levels of human development across countries, based on indicators of health (life expectancy), education, and standard of living (income per capita).

⁷ **Bank Stability Index (BSI)**: A quantitative measure of riverbank stability based on intrinsic geomechanical and geometric properties that indicate the threshold conditions for failure. In this study, it is represented by the critical volume, the minimum volume of water required to infiltrate the riverbank and trigger failure, where lower values correspond to greater instability and higher susceptibility to failure under hydrological loading.

In this study, the BSI is interpreted as the ‘minimum volume of water required to infiltrate the riverbank and trigger failure’. While this is an intrinsic measure, determined by the GGP of the riverbank, it is also indirectly influenced by extrinsic factors, as the time required to reach this volume depends on the rate of runoff generation. This allows intrinsic riverbank behaviour to be analysed alongside extrinsic drivers.

Research Question: How can urbanisation-induced changes in impervious surface area be used to model runoff in the Mula-Mutha river system, and how can intrinsic riverbank properties be combined to introduce a Bank Stability Index (BSI) that can be integrated with runoff to determine time to riverbank failure?

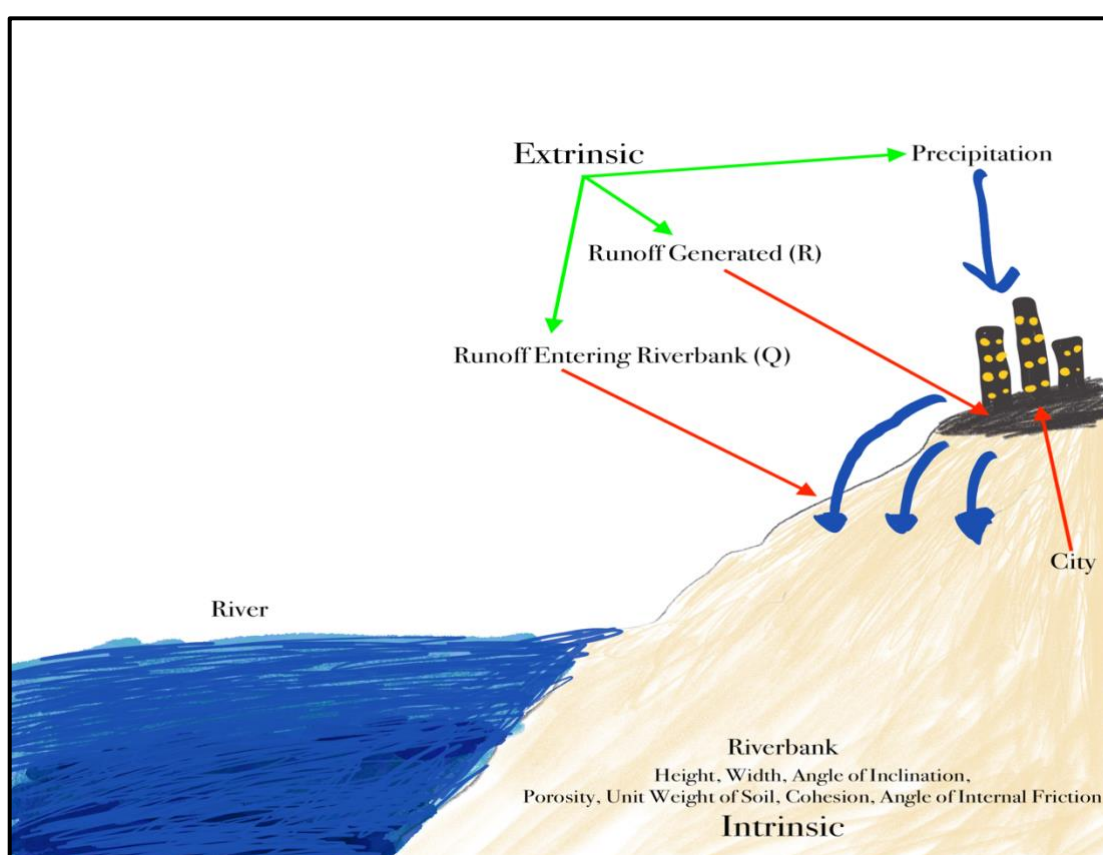


Figure: Pictorial overview of extrinsic and intrinsic factors.

Factor	Sub-factors	Variables Influenced	Research Method/Modelling Approach ⁸	Report Sections	
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⁸ For a more thorough overview of the research methodologies used to answer the research question, refer to Appendix K.

Extrinsic	Urbanisation (Primary Extrinsic Factor)	Impervious Surface Area (ISA) and Non-Impervious Surface Area (NISA)	Google Earth Pro (GEP) manual delineation of ISAs and Google Earth Engine (GEE) for more precise classification.	1.1 , 1.2 , 1.3 , 2	Integrated to calculate R .	Integrated to calculate critical time.
		Rate of Surface Runoff Generation (R)	Rational Runoff Model (RRM) along with ISA estimates.	1.4 , 1.5		
	Precipitation Intensity	Rate of Surface Runoff Generation (R)	RRM, when rainfall intensity is constant and uniform, or incorporated via Modified Rational Runoff Model (MRRM), when the intensity varies temporally and spatial	1.4 , 1.5		
	Lag Function	Time of Concentration, Instantaneous Volume of Water Entering River (Q) ⁹	MRRM.	1.5		
	Intrinsic	Geometric	Height	These factors were shortlisted as they are used to apply the Mohr-Coulomb Failure Criterion (MCFC) to the riverbank, modelled as an inclined plane. Reasons for choosing the MCFC over other constitutive models are provided in 3 .	3	
Angle of Inclination						
Width						
Geomechanical		Porosity				
		Cohesion				
		Angle of internal friction				
		Unit weight of soil				
		Hydraulic conductivity	In the riverbank stability model used to develop the BSI, hydraulic conductivity is assumed to be infinite for simplicity and to enable application of the MCFC.	3 , 4		

Table: *Extrinsic and Intrinsic Factors Affecting Flood Risk.*

1 Urbanisation and Runoff Response

1.1 Study Area and Assumptions

We examine runoff as an urbanisation-driven factor, increased urbanisation raises runoff and peak river flow, heightening overtopping risk.

⁹ It should be noted that the rate of surface runoff generation (R), and thus urbanisation, influences the instantaneous volume of runoff entering (Q) the river. However, only lag time affects Q and not R . Therefore, Q appears only under R .

For the purpose of this analysis, we assume that [all runoff generated eventually enters the river \(A2.1\)](#).¹⁰¹¹ Before introducing impervious and non-impervious surface areas, we define the region for which this urbanisation-induced flood risk pilot study was conducted, shown in **Figure2**.

The study area is Baner-Balewadi Region (BBR) in western Pune, bounded by Pashan Hill (S-SE), National Highway 48 (W), and Mula River (N-E), covering 8.62 km² (8620731 m²).

1.2 Impervious and Non-Impervious Surfaces

Impervious Surface Area (ISA) denotes land surfaces that prevent water infiltration, increasing surface runoff [AG96], such as roads, buildings and parking lots.

Non-Impervious Surface Area (NISA) comprises land allowing water infiltration, generating no surface runoff, such as forests, slopes, and agricultural land.¹²

Different land uses generate varying amounts of surface runoff even within ISA types, such as driveways and roofs. To quantify this, a numerical quantity called the **Runoff Coefficient**¹³ (RC) is assigned to each land use. RC values are typically experimentally determined; in this study, we use data recorded by the Maine Department of Environmental Protection¹⁴ (MDEP).

We assume [a single RC for all ISAs \(\$C_{ISA} = 0.84\$ \) and all NISAs \(\$C_{NISA} = 0.30\$ \) \(A2.2\)](#), to avoid excessive classification complexity. However, this introduces a source of error [\(1.5\)](#).

The methodology used to select the values of C_{ISA} and C_{NISA} is described below.

From **Table1**, the primary ISA land uses relevant for the BBR are ‘Asphalt and Concrete’ and ‘Roof’. The BBR does not contain significant ‘Industrial’ land use, and the ‘Residential’ category can be further subdivided into ‘Roof’ and ‘Asphalt and Concrete’. The RC for ‘Asphalt and Concrete’ ranges from 0.70 to 0.95; we adopt a value of

¹⁰ Assumptions are denoted in blue and assigned a reference number for subsequent reference.

¹¹ This assumption is revisited in [Section 1.5](#).

¹² See Appendix E for typical examples of ISA and NISA used in this study.

¹³ **Runoff Coefficient (RC)**: The volume of surface runoff divided by the volume of rainfall.

¹⁴ The Maine Department of Environmental Protection is a state agency responsible for protecting and restoring Maine's natural resources, air, water, and land, to ensure a healthy environment.

$$C_{\text{Asphalt\&Concrete}} = \frac{0.70 + 0.95}{2} = 0.8375.$$

calculated as the **arithmetic mean**¹⁵ of the minimum and maximum. In the absence of site specific data, the mean is a common pragmatic approach to estimate a representative value. Similarly,

$$C_{\text{Roof}} = \frac{0.75 + 0.95}{2} = 0.85.$$

The mean of C_{Roof} and $C_{\text{Asphalt\&Concrete}}$ gives a representative value for C_{ISA} , assuming equal contributions from each land use type (A2.2.1).

$$C_{\text{ISA}} = \frac{C_{\text{Roof}} + C_{\text{Asphalt\&Concrete}}}{2} = \frac{0.8375 + 0.85}{2} = 0.84375 \approx 0.84.$$

For more detailed studies, area-weighted averaging could be applied if land-use fractions are known. We have rounded the value of C_{ISA} to two **decimal places**¹⁶, which ensures consistency with the RCs in **Table1**.

Similarly, we calculate a representative value for C_{NISA} . The NISA land uses relevant are ‘Heavy Soil, Flat (2%)’ and ‘Heavy Soil, no Crop’. The RC for ‘Heavy Soil, Flat (2%)’ ranges from 0.13 to 0.17 so

$$C_{\text{HeavySoil,Flat(2\%)}} = \frac{0.13 + 0.17}{2} = 0.15.$$

Similarly,

$$C_{\text{HeavySoil,Flat(2\%)}} = \frac{0.30 + 0.60}{2} = 0.45.$$

Taking the mean of $C_{\text{HeavySoil,Flat (2\%)}}$ and $C_{\text{HeavySoil,NoCrop}}$ gives a representative value for C_{NISA}

$$C_{\text{NISA}} = \frac{0.15 + 0.45}{2} = 0.30.$$

¹⁵ **Arithmetic mean:** a measure of central tendency calculated by summing all values in a dataset and dividing by the number of values.

¹⁶ **Decimal places:** the number of digits that appear to the right of the decimal point in a number.

1.3 Temporal Changes in ISA and NISA

We track the NISA and ISA for March 2013, February 2014, and May 2023 to capture short-term variability and longer-term land-use changes, while ensuring the use of clear, high-quality satellite imagery for consistent comparison. The data from historic satellite imagery via GEP, which is summarized in **Table2**, and in **Figure3**.

ISA increased from 2013 to 2023 with an average annual percentage increase of 1.80%, which is notably lower than the sharp 3.05% increase observed between 2013 and 2014.¹⁷

1.4 Applying the Rational Runoff Model

Using **Table2**, expected **surface runoff generation rates**¹⁸ (R), are calculated using the Rational Runoff Model (RRM):

$$R = CAI.$$

Here, C is the RC, A is the area and I is the rainfall's **intensity**¹⁹, assumed to be [uniformly distributed across \$A\$ \(A2.3\)](#).

This expression must be tweaked when A is subdivided into regions A_{ISA} and A_{NISA} , each with *different* RCs C_{ISA} and C_{NISA} ; we replace C with the weighted RC C_{weighted}

$$C_{\text{weighted}} = \frac{C_{ISA}A_{ISA} + C_{NISA}A_{NISA}}{A}.$$

A direct consequence is $R = C_{\text{weighted}}AI$; the formula for C_{weighted} can be extended to deal with situations where there are more than 2 regions ([1.5](#)).

Following the American Meteorological Society, [Gli00], we divide rainfall into categories based on its intensity:

- Light ($< 2.5\text{mm/hr}$),
- moderate ($2.5\text{mm/hr} < I < 7.6\text{mm/hr}$), and
- heavy ($7.6\text{mm/hr} < I < 50\text{mm/hr}$).

¹⁷ Throughout the period, the greatest concentration of NISA has been in the north, toward the Mula River, and in the south, near Pashan Hill. Visual evidence, and the fact that the average annual percentage ISA growth from 2013 to 2014 is less than the same figure from 2013 to 2014, suggests that the rate of development along to Baner Road (central Baner) has slowed over time, likely due to the area approaching saturation.

¹⁸ **Surface runoff generation rate:** the rate at which rainfall is converted into surface runoff over a given area, typically expressed as volume per unit time.

¹⁹ **Intensity** (of rainfall): the rate at which rain falls over a specific area, usually expressed in millimetres or inches per hour.

Table3 calculates C_{weighted} for the three years under study. Next, we calculate R for the three precipitation intensities; results are summarised in **Table4**.

According to the calculations, R has increased from 2013 to 2023. For the upper bound (UB) of light rainfall (2.5mm/hr), there has been a 19.02% increase in R over the period, an 18.94% increase for moderate rainfall UB and an 18.95% increase for heavy rainfall UB. Under [A2.1](#), there is a clear increase in the volume of water flowing in the Mula River following precipitation.

1.5 Limitations and Improvisations

We now highlight the key limitations of our methodology for calculating runoff and discuss possible approaches to address them; these limitations are both practical and rooted in the theoretical framework.

First, in the abstract case, it may not always be possible to partition a region A into subregions A_i such that each A_i has a constant runoff coefficient C_i , since the RC may vary continuously from point to point. In that case, we introduce a **function**²⁰ $q : A \rightarrow \mathbb{R}$, where $q(x, y)$ is the RC at the point $(x, y) \in A$.

Drawing on calculus, we can construct a method to determine the effective RC (C) for this setup. Partition A into areas A_i , with C_i the RC of some point in A_i ; we estimate C by

$$C \approx \frac{1}{A} \sum_i A_i C_i.$$

Taking the **limit**, we obtain

$$C = \frac{1}{A} \int_A q(x, y) dx dy.$$

This is a theoretical derivation and is difficult to implement in practise as determining the function q explicitly is challenging.²¹

²⁰ **Function:** A rule that assigns each input exactly one output, used to describe relationships between variables in Mathematics.

²¹ More complex equations are inserted as images rather than typed directly in Word to ensure consistency and prevent corruption across different Word versions.

Instead of proceeding directly to this framework, we may retain representative values for C_{ISA} and C_{NISA} , as in 1.3. However, we previously used an **unweighted average**²² which may not be representative if ISAs have significantly different proportions (A.2.1). In such cases a **weighted average**²³ is more appropriate, given by

$$C_{ISA} = \frac{\sum_{i=1}^n C_{ISA}^i A_{ISA}^i}{\sum_{i=1}^n A_{ISA}^i}.$$

Here the A_{ISA}^i from $i = 1$ through to $i = n$ are the distinct ISAs and C_{ISA}^i is its RC. Note that $A_{ISA} = \sum_{i=1}^n A_{ISA}^i$. Similarly,

$$C_{NISA} = \frac{\sum_{i=1}^m C_{NISA}^i A_{NISA}^i}{\sum_{i=1}^m A_{NISA}^i}.$$

As noted in 1.3, these formulae require knowing the areas of different ISAs and NISAs, which is tedious.

Furthermore, the success of the RRM depends on the accuracy of the RCs, and we could not experimentally determine them for the specific surface types in the BBR.²⁴ The actual RCs may differ from those in the MDEP database, which is based on U.S. conditions.

Another limitation lies in the validity of the RRM itself. We assumed (following the model's hypothesis) uniform rainfall intensity (A2.3); however, in practise rainfall is rarely uniform. Localised storms may cause some regions to receive more rainfall than others. It is also possible that regions receiving the highest rainfall have the longer flow paths to the river, reducing immediate flood risk.

We use this opportunity to relax A2.3 and, at the same time, modify A2.1; we begin by defining functions i and τ .

- **Intensity:** $i : A \times [0, \infty) \rightarrow [0, \infty)$ where $i(x, y, t)$ is the rainfall intensity at $(x, y) \in A$ and at time t (relaxes A2.3). This allows rainfall to vary both spatially and temporally.
- **Time Lag:** $\tau : A \rightarrow [0, \infty]$ where $\tau(x, y)$ is the time taken by a droplet of water, starting from $(x, y) \in A$ to reach the closest point in the river. The **time of concentration**²⁵ (t_c), a property of a drainage basin, is related to τ by

²² **Unweighted Average:** Average where all values are equal; sum of values $\div$ number of values.

²³ **Weighted Average:** Average where values have different importance; sum of (value $\times$ weight) $\div$ sum of weights.

²⁴ In our notation, the RCs here are C_{ISA}^i and C_{NISA}^i .

²⁵ **Time of concentration:** The time required for water to travel from the most distant point of a drainage area to its outlet.

$$t_c = \max_{(x,y) \in A} \tau(x,y).$$

- In defining τ , a single pathway from (x,y) to the river is assumed. This idealisation is partially justified in urban environments where runoff tends to follow well-defined drainage paths; flow is more likely to spread, bifurcate, or experience temporary storage in rural and open environments.
- We also allow τ to take the value ∞ , representing regions where runoff cannot reach the river due to obstructions thereby relaxing [A2.1](#). In this case $i(x,y, -\infty) = 0$.

The introduction of τ allows us to define $Q(t)$, the **instantaneous volume of runoff entering the river**²⁶, which will be useful when integrating extrinsic and intrinsic factors in [3.6](#).

- For an **infinitesimal**²⁷ element dA , the contribution to river flow is

$$dQ = q(x,y)i(x,y,t - \tau(x,y))dA.$$

- If $\tau(x,y) = \infty$, then $dQ = 0$, which is physically consistent.
- Integrating over entire area A gives

$$Q(t) = \int_A q(x,y)i(x,y,t - \tau(x,y))dxdy.$$

To see the time of concentration in action, we state Q in **piecewise**²⁸ form:

$$Q(t) = \begin{cases} 0 & \text{if } t < \min_{(x,y) \in A} \tau(x,y) \\ \int_{A_t} q(x,y)i(x,y,t - \tau(x,y))dxdy & \text{if } \min_{(x,y) \in A} \tau(x,y) \leq t \leq t_c \\ \int_A q(x,y)i(x,y,t - \tau(x,y))dxdy & \text{if } t > t_c \end{cases}$$

where

$$A_t = \{(x,y) \in A : \tau(x,y) \leq t\}.$$

²⁶ **Instantaneous volume of runoff entering river:** The volume of surface runoff entering a river at the given time.

²⁷ **Infinitesimal:** An extremely small quantity closer to zero than any real number, used in calculus to describe continuous change in Calculus.

²⁸ **Piecewise function:** A function defined by different expressions for different intervals of its domain.

This shows how properties of the drainage basin may affect the river's response to runoff. For instance, if $\min_{(x,y) \in A} \tau(x,y)$ is large, then the river will take long to respond to the precipitation event.²⁹

We analyse the second case of the piecewise definition as well; let $\min_{(x,y) \in A} \tau(x,y) \leq t \leq t_c$. Using the **ML inequality**³⁰ to bound $Q(t)$

$$Q(t) = \int_{A_t} q(x,y) i(x,y, t - \tau(x,y)) dx dy \leq \text{Area}(A_t) \max_{(x,y) \in A_t} q(x,y) i(x,y, t - \tau(x,y)).$$

We can also bound Q from below:

$$\text{Area}(A_t) q_{\min}(t) i_{\min}(t) \leq Q(t) \leq \text{Area}(A_t) q_{\max}(t) i_{\max}(t), \text{ for } \min_{(x,y) \in A} \tau(x,y) \leq t \leq t_c.$$

Here, $q_{\min}(t) = \min_{(x,y) \in A_t} q(x,y)$, $q_{\max}(t) = \max_{(x,y) \in A_t} q(x,y)$, with $i_{\min}(t)$ and $i_{\max}(t)$ defined similarly.

These inequalities show that the larger $\text{Area}(A_t)$, the larger the volume of runoff deposited in the river at time t . Note that $\text{Area}(A_t)$ is a factor dependent on the geometry of the drainage basin.

²⁹ Q will remain 0 for a longer period of time initially.

³⁰ **ML inequality:** The absolute value of a double integral can't be bigger than the largest value of the function times the area of the region.

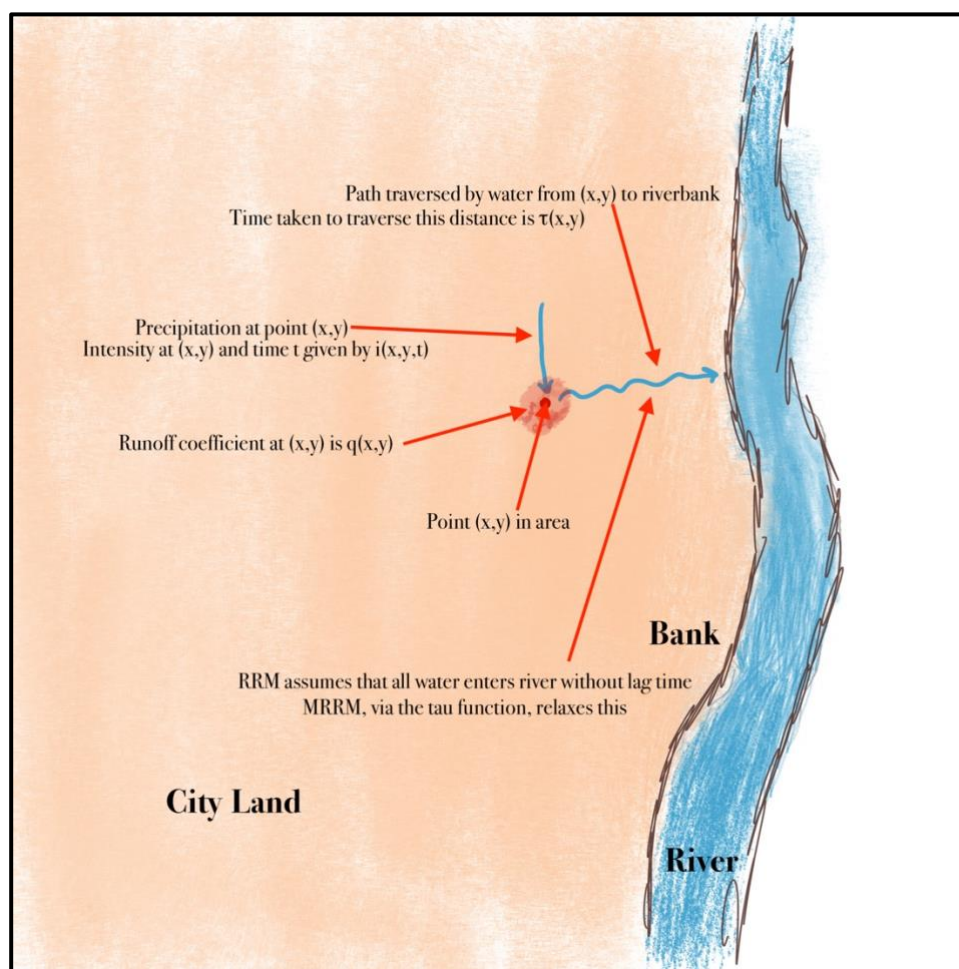


Figure: Pictorial representation of the MRRM.

This formulation, the Modified Rational Runoff Model (MRRM), does generalise the RRM: if $q(x,y)$ and $i(x,y,t)$ are assumed to be constant, and $\tau(x,y)$ is negligible, the expression reduces to the standard $R = CAI$. Moreover, the MRRM is theoretically more accurate, in that it distinguishes between runoff generated (R) and runoff entering the river (Q), as only the latter directly affects flood vulnerability (3.6), and hence is more relevant in this study. However, its practical implementation is limited by the difficulty of determining the functions $q(x,y)$, $i(x,y,t)$, and $\tau(x,y)$.³¹

Additionally, both the RRM and MRRM assume that RC does *not* vary with time. In reality, as surfaces become saturated infiltration decreases, potentially increasing runoff.³²

³¹ Note that we will use values of R instead of Q when integrating with the critical volume BSI (Section 3.6), as Q , while theoretically formulated above, cannot be calculated due to the lack of explicit function for τ . That is, A2.1 will remain in place.

³² That is, the function $q(x,y,t)$ is such that $\partial_t q(x,y,t) > 0$.

2 Remote Sensing for Urban Surface Mapping

2.1 Manual Mapping Limitations

One limitation in the method used in [1](#) is the manual delineation of NISA boundaries; this introduces subjectivity and classification error, as regions may be omitted or misclassified. To reduce subjectivity, we automate the classification process to determine A_{ISA} and A_{NISA} .

Remote sensing measures the physical characteristics of an area via its reflected and emitted **electromagnetic radiation**³³ remotely, typically using satellites. For a more complete introduction to remote sensing, see Appendix H.

2.2 NDVI for Surface Classification

The main **index** that we consider is the **normalized difference vegetation index**³⁴ (NDVI). NDVI has desirable properties, as described in **Table5** [Gao96].

Thus, NDVI is used to distinguish between ISA and NISA: if the NDVI value is below 0.3, assuming no water or clouds present, the area is impervious; this condition is used to calculate A_{ISA} . The spatial resolution of Sentinel-2 is 10m, limiting the accuracy of this process, as many structures, including parks, buildings, and homes, have areas smaller than 100m² and may not be resolved within a single pixel. Higher-resolution data wasn't freely available, and the cost of acquiring meter-resolution satellite imagery for the area was high. Nevertheless, this provided significantly improved spatial detail than Landsat, the only other freely available option, which, while offering greater temporal coverage since 1970, has a lower resolution of 30m.

2.3 Google Earth Engine (GEE) Implementation

We use GEE to compute the area within the BBR where $NDVI < 0.3$.³⁵ Tutorials by Spatial Thoughts guided our use of the GEE API and GEE datasets.

³³ **Electromagnetic radiation:** Energy that travels through space in the form of waves, such as light, radio waves, and X-rays.

³⁴ **Normalized Difference Vegetation Index:** A measure of vegetation health calculated from satellite data by comparing near-infrared and red light; higher values indicate denser, healthier vegetation.

³⁵ See **Code1** in Appendix E for the GEE script implementing this logic.

As NDVI thresholds are context-dependent, we assess sensitivity by increasing the threshold from 0.15 to 0.30 in increments of 0.01. The lower the threshold, the less strict the classification becomes: we expect NISA to increase and the ISA to decrease. Results with a threshold of 0.30 are summarised in **Table6**, with graphs for multiple thresholds in **Figure4**.

We focus on the period from 1st January to 30th April, when conditions are typically dry in Pune.³⁶ A more precise approach would filter out days with precipitation. However, this is not entirely reliable, as sunny days may still have residual moisture from previous rainfall.

The NDVI based analysis produced counterintuitive results, indicating a decreasing trend in total ISA in the BBR from 2017 to 2023 across all threshold values. Several factors may have contributed to this pattern.

- NDVI, by definition, is sensitive to vegetation rather than impervious surfaces, meaning that small increases in trees, lawns, or roadside greenery within urban pixels can elevate NDVI values.
- The 10m spatial resolution of Sentinel-2 leads to mixed pixels containing both impervious surfaces and vegetation; even minor vegetation coverage within a pixel can shift it above the impervious threshold.

Collectively, these limitations suggest that NDVI may not fully capture changes in urban surface cover, motivating the use of alternative indices.

2.4 Beyond NDVI

NDVI is designed to measure vegetation, not impervious surfaces. Some impervious surfaces may show moderate NDVI values due to dust or moisture.

We therefore test other indices, including the **NDBaI**³⁷. In contrast to NDVI, which is an older and less specialized index, NDBaI is a more recently introduced index designed to specifically distinguish between ISAs and bare soil [ZCo5]. It is therefore an alternative to NDVI for non-vegetated areas. As with NDVI, it ranges from -1.0 to 1.0. The properties that led us to consider this index are listed in **Table7**.

³⁶ The wet season occurs from June to September.

³⁷ This is an abbreviation for Normalized Difference Bareness Index.

Figure5 shows ISA percentage in the BBR from 2017 to 2026; with an overall upward, though not strictly increasing, trend throughout the period under study, suggesting NDBaI may better capture ISA.

While remote sensing, at least in theory, can reduce subjectivity, result accuracy depends on the choice of index; the contrasting trends from NDVI and NDBaI highlight that no single index is reliable across all environments. Therefore, experimenting with multiple indices for the study area and objective is necessary.

3 Riverbank Stability Modelling

3.1 Runoff and Riverbank Stability

Earlier sections quantified the impact of urbanization on flood risk through changes in ISA and runoff, but this does not capture the river channel's physical response, which depends on the GGP's of the riverbank.

Runoff increases discharge, raising the erosive force of the water and promoting riverbank **erosion**³⁸ [Kni14]. However, while erosion acts externally on the riverbank surface, riverbank failure may occur internally due to increases in **pore-water pressure**³⁹ (u) and corresponding reductions in **effective normal stress**⁴⁰ (σ_{eff}) caused by runoff infiltration. If such failure leads to collapse of riverbank, the river's **conveyance capacity**⁴¹ is reduced due to increased **sediment load**, thereby increasing the likelihood of flooding [Jul10].

3.2 Bank Geometry and Assumptions

To quantify bank vulnerability, the minimum volume of runoff required to trigger **failure**, the **critical volume** (V_{crit}), is estimated. This matters because lower values of V_{crit} imply a more unstable riverbank, simply because the time to failure, the **critical time** (t_{crit}), is also lower; conversely, greater V_{crit} values translate to a more stable riverbank, as t_{crit} will be greater.

³⁸ **Erosion**: The gradual wearing away and transport of soil or rock by water, wind, ice, or gravity.

³⁹ **Pore-water pressure**: Pressure exerted by water within the pores of a soil, which reduces the soil's effective strength.

⁴⁰ **Effective normal stress**: Portion of total stress in a soil that is carried by the soil skeleton, equal to total stress minus pore-water pressure.

⁴¹ **Conveyance capacity**: The maximum amount of water a channel can carry without overflowing.

Taking this a step further, we assume that primary source of water entering a riverbank is runoff, which gives the mathematical relationship

$$\int_0^{t_{\text{crit}}} Q(t)dt = V_{\text{crit}}.$$

By the definition of Q , $\int_0^T Q(t)dt$ is the total volume of water infiltrated into the riverbank at time T after precipitation began. The caveat here is that while Q is expressible in terms of input variables (q , i , τ), the lack of an explicit definition for τ means that solving for t_{crit} is not possible (1.5). Instead, we restore 42.1 and follow the simplified analysis of 1.4, so the expression becomes:

$$\int_0^{t_{\text{crit}}} Q(t)dt = t_{\text{crit}}R = V_{\text{crit}} \Rightarrow t_{\text{crit}} = \frac{V_{\text{crit}}}{R}.$$

This formulation, while less precise, is practical as we've calculated R values previously (**Table4**). The simplified expression for t_{crit} numerically integrates intrinsic and extrinsic factors: lower critical volume (intrinsic) OR greater rate of runoff generation (extrinsic) implies that the bank is more susceptible to failure.

To relate V_{crit} to the GGPs of the riverbank, we define an idealized geometric and hydrological model of it.

- We model the riverbank as an inclined plane with **angle of inclination**⁴² (β), width (b), and height (H), capturing key geometry while keeping the analysis tractable.⁴³
- In addition to geometric variables, we introduce geomechanical variables, including the soil **porosity**⁴⁴ (n), **cohesion**⁴⁵ (c), and **angle of internal friction**⁴⁶ (ϕ). Furthermore, infinite **hydraulic conductivity**⁴⁷ (K) is assumed, so water

⁴² **Angle of inclination**: Angle between an inclined plane and the horizontal surface, indicating how steep the plane is.

⁴³ Indeed, riverbanks are not perfectly planar, as shown in the dataset in the Appendix.

⁴⁴ **Porosity**: Proportion of void (empty) spaces within a soil or material compared to its total volume, indicating its capacity to store water or air.

⁴⁵ **Cohesion**: Internal shear strength of a soil due to attractive forces between particles, which helps the soil resist deformation and failure even in the absence of external pressure.

⁴⁶ **Angle of internal friction**: Measure of a soil's resistance to sliding, representing the frictional strength between particles when shear stress is applied.

⁴⁷ **Hydraulic conductivity**: A measure of a material's ability to allow water to flow through it, typically expressed in meters per second or centimetres per second.

redistributes instantaneously. This implies that water infiltrates upward from the base, creating a triangular cross-section. Moreover, we assume that water cannot percolate beneath the riverbank into the groundwater, reflecting the slow rate at which it reaches the **water table**⁴⁸.

With the physical model defined, we now determine the condition under which failure occurs.

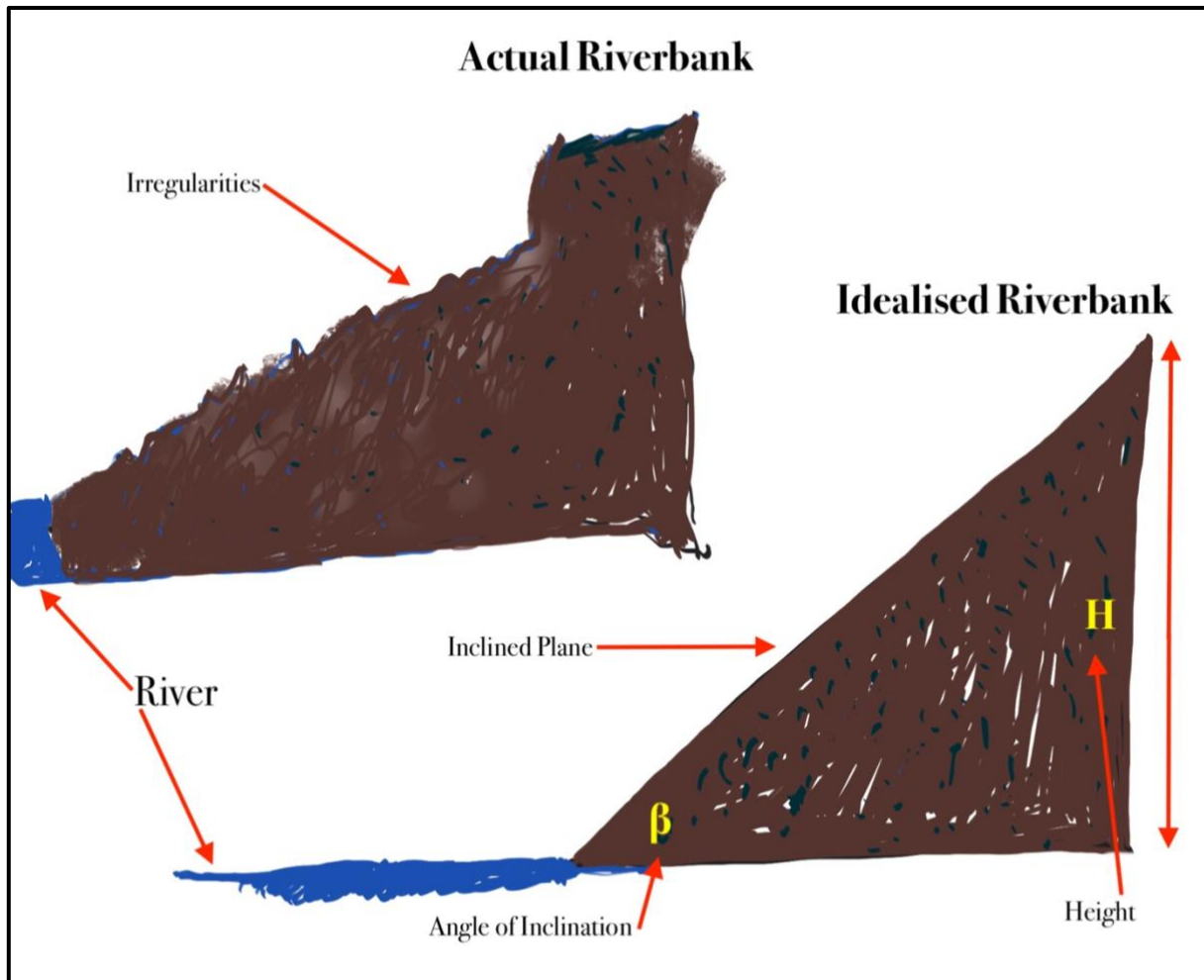


Figure: Pictorial representation of riverbank.

3.3 Mohr-Coulomb Failure Criterion

To determine failure, we apply the **Mohr-Coulomb Failure Criterion** (MCFC), a **constitutive model**, which relates shear stress to effective normal stress and allows stresses to be expressed in terms of geometric variables [Cou76, Moh00]. See Appendix F.

⁴⁸ **Water table:** The upper surface of groundwater below which soil or rock is fully saturated with water.

In our setup, the failure surface is taken as a plane within the riverbank, parallel to the upper surface. As stresses vary along this plane, the MCFC must be applied locally to each point.⁴⁹ However, MCFC does not explicitly account for runoff-induced pore water.

This is incorporated using **Terzaghi's principle** [Ter25, Ter43] ($\sigma_{\text{eff}} = \sigma - u$), where u reduces σ_{eff} , linking runoff to internal stability:

$$\tau_f = c + (\sigma - u) \tan \phi,$$

where τ_f is the **critical shear strength**⁵⁰.

While other constitutive models, such as the **Modified Cam-Clay Model**⁵¹ (MCCM) and **Drucker-Prager Criterion**⁵² (DPC), capture additional soil behaviour, including **strain hardening**⁵³ and the influence of the **intermediate principal stress**⁵⁴, they introduce nonlinearities and thus require **numerical solution methods**⁵⁵ [Son20]. The MCCM is effective for predicting the stress-strain behaviour of clays but involves parameters that typically necessitate **finite element analysis**⁵⁶ [Con12]. Similarly, while the DPC offers a smooth, three-dimensional approximation of soil failure, its associated flow rules can sometimes lead to excessive **dilatancy predictions**⁵⁷ [Son20].

As this study aims to derive a closed-form expression for the critical volume, and hence the BSI, the MCFC provides an appropriate balance between physical reality and computational efficiency. The MCFC remains the 'best known' criterion in soil mechanics due to its linear

⁴⁹ That is, it must be applied on an infinitesimal element.

⁵⁰ **Critical shear strength:** The maximum shear stress that a material (such as soil) can withstand before failure occurs along a shear plane.

⁵¹ **Modified Cam-Clay Model:** A soil model used to describe how clay behaves under loading, especially accounting for plastic deformation and consolidation.

⁵² **Drucker-Prager Criterion:** A yield condition used to predict when materials (like soil or rock) begin to fail, based on overall stress rather than specific planes.

⁵³ **Strain hardening:** the increase in a material's strength and resistance to deformation as it is plastically deformed.

⁵⁴ **Intermediate principal stress:** The second-largest principal stress acting at a point, between the maximum and minimum principal stresses.

⁵⁵ **Numerical solution methods:** Techniques that use approximate calculations (often via computers) to solve mathematical problems that are difficult or impossible to solve exactly.

⁵⁶ **Finite Element Analysis:** A computational method that divides a complex system into small elements to analyse stresses, strains, or other physical behaviours.

⁵⁷ **Dilatancy predictions:** estimates of how much a soil or granular material will expand in volume when it is sheared, reflecting the tendency of particles to move apart during deformation.

relationship between shear and normal stress, allowing for the derivation of analytical solutions, often unattainable with complex, **state-dependent models** [Son20].

3.4 Expression for the Critical Volume

To specialize this to our scenario, we must express σ and τ in terms of H, z (elevation of the point) and β, u in terms of V, n, b and β ; the expressions and derivations are in Appendix G. The expressions for σ and τ show that as β increases, τ increases while σ decreases. Thus, at least without considering water at first, one might tentatively conclude that the steeper the bank, the less the soil is clamped together, and the more unstable it may be.

We substitute the expressions into the MCFC, and set equality to get $V_{\text{crit}}(z)$; the minimum volume of water required to initiate failure at elevation z , i.e. the threshold at which the failure condition is first satisfied locally.⁵⁸ The function $V_{\text{crit}}(z)$ has minimum value 0, occurring at:

$$z_{\text{crit}} = -\frac{c \cot \phi + H\gamma E}{\gamma_w - \gamma E}.$$

However, this value is not always physically admissible: not all slopes fail spontaneously without water, and that failure may instead require a positive accumulation of pore water pressure.⁵⁹

Thus, failure occurs without any runoff addition if and only if

$$\beta > \phi, \text{ and } H \geq \frac{c \cot \phi}{\gamma|E|}.$$

This condition states that the bank must be both steep and tall enough to fail without increase in water content: the slope angle exceeds the soil's internal friction angle, reducing shear resistance, while the height is large enough relative to the soil cohesion that the material can no longer support itself, leading to instability.⁶⁰ Otherwise:

$$V_{\text{crit}}(0) = \frac{nb}{2 \tan \beta} \left(\frac{c \cot \phi}{\gamma_w} + H \frac{\gamma}{\gamma_w} E \right)^2 \text{ and } V_{\text{crit}}(H) = \frac{nb}{2 \tan \beta} \left(\frac{c \cot \phi}{\gamma_w} + H \right)^2.$$

To determine which governs failure, we compare:

$$V_{\text{crit}} = \min\{V_{\text{crit}}(0), V_{\text{crit}}(H)\}.$$

⁵⁸ See Appendix G for the derivation.

⁵⁹ That is, the value of z_{crit} may lie outside the interval $[0, H]$.

⁶⁰ Essentially, $\beta > \phi$ and $H > \frac{c \cot \phi}{\gamma|E|}$ are the conditions required for a **dry landslide** (the downslope movement of soil or rock under gravity without significant involvement of water) to occur.

This shows that the location of failure depends on both geomechanical and geometric properties and must be determined by evaluating both boundary conditions.

3.5 Relationship between GGP and Critical Volume

To simplify the discussion, assume $V_{\text{crit}} = V_{\text{crit}}(0)$; the case where $V_{\text{crit}} = V_{\text{crit}}(H)$ can be handled similarly.

- **V_{crit} is directly proportional to porosity:** more compacted soil has lower porosity, so a smaller volume of water is needed to trigger collapse, meaning the bank may fail more quickly.
- **Cohesion enters inside the squared term, so V_{crit} has an approximately quadratic dependence on c .** This suggests that increasing cohesion significantly raises the volume of water required to trigger failure; for example, a doubling of c may lead to a more than proportional (roughly fourfold, if other terms are small) increase in V_{crit} . Physically, higher cohesion strengthens the soil, making collapse less likely unless a larger volume of water is present.
- **For $0 < \phi < 45^\circ$, $\frac{dV_{\text{crit}}}{d\beta} < 0$, meaning that a steeper bank decreases the volume of water required, reflecting increased instability.** However, note the derivative is neither constant nor linear, reflecting the complex relationship between bank angle and stability, a relationship that initially appeared much simpler before detailed analysis.

V_{crit} integrates multiple geomechanical and geometric factors into a single expression and may therefore be interpreted as a BSI, where lower values correspond to increased riverbank instability. Furthermore, V_{crit} is directly related Q , establishing a link between extrinsic drivers and intrinsic properties, to be made explicit via t_{crit} .⁶¹

3.6 Explicit Calculation of Critical Volume and Time for Mula-Mutha

⁶¹ Recall the relationship $V(t) = \int_0^t Q(t')dt'$. Therefore, the time taken for the critical volume to be achieved is dependent on Q , which itself is dependent on several extrinsic factors, such as rainfall intensity (i) and runoff coefficients (q), outlined in the MRRM ([Section 1.5](#)).

To demonstrate the applicability of V_{crit} as a BSI, we consider parameter values for the Mula-Mutha riverbanks, using the data from the Indian Standard for Deccan Trap Soils [Sin72] (**Table8**).

Based on these values, $V_{crit} = 10.80\text{m}^3/\text{m}^{6263}$; when integrated with the rate of runoff generation from May 2023 for the 5.4km study stretch of the BBR, the model illustrates a decline in t_{crit} with increasing intensity: Under light precipitation, $t_{crit} = 3.92\text{hr}$. However, for moderate precipitation, this window narrows to 1.92hr, and under heavy conditions, the threshold is breached in just 11.7min.

This suggests that under moderate to heavy rainfall, the runoff generated may exceed the stability threshold within minutes to hours, indicating a heightened risk of bank failure. While approximate, these calculations demonstrate how the BSI can be used as a practical indicator of flood vulnerability.

To reiterate, we used values of R instead of Q when calculating the time required for reach critical volume, effectively assuming that $R = Q$ (all runoff enters the river, [A2.1](#)) due to the lack of explicit function for τ . Since $Q \leq R$, substituting R for Q yields $t_{crit} = \frac{V_{crit}}{R} \leq \frac{V_{crit}}{Q}$. Therefore, the model underestimates the critical time, predicting earlier failure than would occur in reality.

⁶² As with all other data points, is V_{crit} given to 2 decimal places.

⁶³ We have divided the raw critical volume by the length of the river channel, to get a normalised critical volume, so that the units are m^3/m .

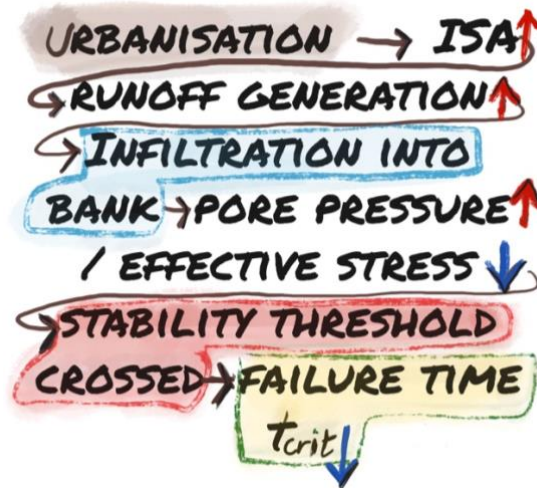


Figure: Flowchart connecting extrinsic factors to intrinsic factors.

Furthermore, this model relies on two simplifying assumptions, limiting its applicability to real-world systems, especially where spatial heterogeneity and finite hydraulic conductivity play significant roles.

3.7 Model Limitations and Localised Variables

1. **Infinite K :** Assuming infinite hydraulic conductivity implies instantaneous redistribution of water, preventing localised pore pressure building. In reality, finite conductivity would lead to accumulation of pore water, increasing u more rapidly and reducing effective stress [Gu14]. Consequently, this assumption likely overestimates stability and underestimates failure risk.
2. **Homogeneous soil properties:** Properties such as porosity are assumed to be uniform throughout the riverbank, allowing for tractable expressions such as V_{crit} . However, natural soils exhibit significant spatial heterogeneity, with properties varying across different scales.

To address the latter limitation, the next section focuses on redefining the Representative Elementary Volume (REV) as a localized, scale-dependent quantity for parameters such as porosity and volumetric water content. This redefinition provides a framework for capturing variability and lays the groundwork for locally varying properties, including hydraulic conductivity.

At present, the analysis is limited to refining the mathematical definition of the REV; a detailed formulation of microscale hydraulic conductivity and its integration into macroscopic stability models is beyond the scope.

4 Redefining the Representative Elementary Volume

Hydraulic conductivity depends on the **volumetric water content** (VWC, θ), defined as the ratio of water volume to total soil volume [Hil98]. Thus, defining hydraulic conductivity **locally** requires defining VWC locally. That is, we must specify the function $\theta(\mathbf{x})$, where $\mathbf{x}$ is a point in the sample.

To address this, we consider a **localised version**⁶⁴ of the standard REV, introduced by [AP10]:

$$\theta_{\text{REV}}(\mathbf{x}) := \inf\{|V_1| : |\theta(\mathbf{x}, V_2) - \theta(\mathbf{x}, V_1)| < \varepsilon \text{ for all } |V_2| > |V_1|\},$$

where ε is a fixed **tolerance**^{65, 66}. Then, we set

$$\theta(\mathbf{x}) = \theta(\mathbf{x}, \theta_{\text{REV}}(\mathbf{x})).$$

Physically, one expects that for small volumes,

$$\theta(\mathbf{x}, V) = \int_{V(\mathbf{x})} \chi_{\text{water}}(\mathbf{x}) dV,$$

fluctuates significantly due to **pore-scale variations**⁶⁷. As the volume increases, these fluctuations diminish, and the value stabilises; this is the reason for calculating the lower bound of $\{|V_1| : |\theta(\mathbf{x}, V_2) - \theta(\mathbf{x}, V_1)| < \varepsilon \text{ for all } |V_2| > |V_1|\}$ in the expression for $\theta_{\text{REV}}(\mathbf{x})$.⁶⁸

⁶⁴ **Localised version:** A formulation of a concept or model that is adapted to apply at a specific point or small region in space, typically by introducing position dependence (e.g., functions of $\mathbf{x}$).

⁶⁵ **Tolerance:** A predefined small value that determines when variations in a measured quantity are considered negligible.

⁶⁶ Throughout, V_1 and V_2 denote volumes around the point $\mathbf{x}$.

⁶⁷ **Pore-scale variations:** Fluctuations in properties caused by the microscopic structure of soil pores and the distribution of water within them.

⁶⁸ Here, the function χ_{water} is defined to be 1 if $\mathbf{x}$ is in a water-containing pore and 0 otherwise. Furthermore, this notion can be extended to other porous material properties. For example, porosity can be defined similarly by replacing χ_{water} with an indicator function χ_{void} . This takes the value 1 in pore space and otherwise.

However, this definition of the REV implicitly assumes homogeneous soil, such that beyond a certain scale, properties become independent of sample size. Indeed, the REV used by [AP10] and [Sch22, Sch24] is one-sided: it does not impose an upper bound on $V(\mathbf{x})$.⁶⁹ While it ensures the REV is larger than the **pore-scale**⁷⁰, it does not require it to remain smaller than the **sample-scale**⁷¹; the condition $|\theta(\mathbf{x}, V_2) - \theta(\mathbf{x}, V_1)| < \varepsilon$ for all $|V_2| > |V_1|$ is only satisfied for homogenous media.

To address this limitation, we redefine the REV by introducing *both* lower and upper bounds: it must be large enough to smooth pore-scale variations, yet small enough to preserve spatial variability across regions: a lower bound (REV_L), corresponding to the transition from the pore-scale (i.e., microscale), and an upper bound (REV_U), corresponding to the onset of sample-scale (i.e., macroscale) heterogeneity is stated in Appendix I; this redefinition ensures that the $\theta(\mathbf{x})$ remains a local property, capturing the variability of the soil.

Conclusion

Our central contribution lies in the development of two modelling constructs bridging urban hydrology (MRRM) and riverbank geomechanics (BSI).

Using Google Earth Pro to obtain values for ISA and NISA, we were able to estimate surface runoff generation rates under varying precipitation intensities. A well-known consequence of this increase in runoff rates due to urbanisation is that nearby rivers, such as the Mula-Mutha in Pune's case, face a greater risk of overtopping. However, we note a more nuanced and much less studied consequence: runoff infiltrates riverbanks and increases pore water pressure, compromising bank stability and enhancing flood risk by reducing conveyance capacity.

To make this causal link precise, we modelled the riverbank as an inclined plane, enabling the application of the MCFC, allowing us derive a novel expression for the minimum volume of water required to trigger collapse. We selected the MCFC over other failure models, such as the MCCM, as they require greater computational power, and hence to do not yield a clean, analytically tractable expression for V_{crit} .

⁶⁹ They define the REV as the minimum volume at which a given parameter becomes independent of sample size.

⁷⁰ **Pore-scale**: The microscopic scale at which individual soil pores, grains, and fluid distributions are resolved, and where properties exhibit strong local variability. Also known as the micro-scale.

⁷¹ **Sample-scale**: The macroscopic scale of the entire soil sample or domain, over which bulk or averaged properties are measured, and spatial heterogeneity may become significant. Also known as the macroscale.

Our BSI approach not only revealed nuanced relationships between GGPs and riverbank stability (for instance, there was no simple relationship between inclination and stability, apart from a *general* inverse trend), but also allowed the formulation of parameters such as t_{crit} which combine extrinsic and intrinsic factors into a single expression, thereby accounting for both urban changes and river infrastructural developments. This integrated our work in [1](#) and [2](#) with that in [3.1](#) to [3.5](#).

Although our novel MRRM introduces lag time, allowing calculation of Q , a significant improvement over conventional RRM, the lack of data for the BBR catchment geometry required using R instead of the more relevant Q values to calculate t_{crit} , leading to its underestimation. We hope to address this limitation, as well as the assumption of homogeneous intrinsic properties when deriving the BSI via our reformulation of the REV, in future work.